replacement.py

`src/moe_pruning/pruning/replacement.py · 176 lines · module moe_pruning.pruning.replacement`

What it does

Removes the selected experts and installs their replacement for the chosen `REPLACEMENT_METHOD`: `bias` (BiasExpert), `drop` (ZeroExpert) or `reroute` (ZeroExpert plus a routing mask).

How it works

- `replace_experts()` builds the replacement modules on each layer's own device, lets the adapter install them, frees the original weights, verifies that router weights are bit-identical and that no original weights of removed experts remain.
- `ReplacementReport` records removed and added parameters and the net reduction.
- `bias_parameters()` and `freeze_all_but_biases()` prepare bias training: everything except BiasExpert vectors is frozen, and the result is verified.

Defined in this file

Extracted from the source code (classes, public functions and methods).

Name	Description
`class ReplacementReport`	Replaced experts and removed / added parameter counts.
`.to_dict()`	Convert to a plain dictionary for JSON reports.
`make_replacement_module(method, output_dim, init, dtype, device)`	Module installed in a removed expert's slot for the given method.
`replace_experts(model, adapter, layers, selection, method, biases)`	Remove the selected experts in place and install their replacement.
`replace_experts_with_biases(model, adapter, layers, biases)`	Shorthand for `replace_experts(..., method="bias")`.
`bias_parameters(model)`	Named parameters belonging to BiasExpert modules.
`freeze_all_but_biases(model)`	Freeze every parameter except BiasExpert.bias and return the trainable ones.

Connections

Uses: `moe_pruning.model.architecture`, `moe_pruning.model.placement`, `moe_pruning.pruning.bias_expert`, `moe_pruning.pruning.zero_expert`

Used by: `scripts/replace_experts.py`, `scripts/train_biases.py`, `src/moe_pruning/serialization/checkpoint.py`